

Roll No.....

SET-II

End Semester Examination, 2025

Name of the Program: B.Tech Biotechnology

Semester: VI

Name of the Course: Bio-separation Technology

Course Code: TBT603

Time: 3 hrs.

Maximum Marks: 100

Note:

(i) Answer any two questions from each section

(ii) Each question carries 10 marks

Q1	(20 marks)	
(a)	Which key steps are typically included in the schematic diagram of a downstream processing workflow, and what is the function of each step?	CO1, CO2
(b)	How is adsorption used in the recovery of bioproducts, and what materials are commonly used?	
(c)	Why is understanding the hydrophobic nature of a molecule important in designing a separation strategy?	
Q2	(20 marks)	
(a)	How does centrifugation work, and what parameters affect its efficiency?	CO2 CO3
(b)	Describe the principle and working mechanism of SDS-PAGE. How does it allow for the separation of proteins based on molecular weight?	
(c)	Explain how cell disruption affects the choice of debris removal technique.	
Q3	(20 marks)	
(a)	How is isoelectric focusing used in protein separation? Discuss 2 D electrophoresis.	CO2,C O3
(b)	How does the settling velocity of particles influence the design of sedimentation tanks in large-scale bioprocessing?	
(c)	Compare gel filtration chromatography and hydrophobic interaction chromatography in terms of their separation mechanisms.	
Q4	(20 marks)	
(a)	Explain the principle and application of crystallization in the purification of bioproducts	CO4, CO5
(b)	Discuss the use of activated sludge systems and membrane bioreactor (MBR) technology in municipal and industrial wastewater treatment.	
(c)	What is aqueous two-phase extraction (ATPE), and how is it useful for protein or enzyme separation?	
Q5	(20 marks)	
(a)	What are the major types of drying methods used in biotechnology, and how do they differ?	CO4, CO5
(b)	Describe the general stages of effluent treatment (primary, secondary, tertiary) in an industrial setting	
(c)	Discuss how improper drying can affect the activity and quality of enzymes or pharmaceuticals.	

